Rafael Moretti-Klein

Backend Engineer | Python, Go

PROFESSIONAL SUMMARY

Backend engineer and former engineering leader with 3+ years of experience building and migrating Python backend services, alongside substantial production Go experience. Work includes high-throughput entitlement services, Python and Go service migration, API reliability, and observability on GCP and Kubernetes. Now focused on hands-on individual-contributor backend engineering.

TECHNICAL SKILLS

Languages: Python, Python 3, Go, C++
Backend and APIs: FastAPI, gRPC, playback API development
Data and storage: Cassandra, Redis, BigQuery
Cloud and infrastructure: GCP, GKE, Kubernetes, Terraform
Observability: OpenTelemetry, Prometheus
Testing: pytest, tox

PROFESSIONAL EXPERIENCE

Streamhaus GmbH *2015–Aug 2026*

Director of Engineering *2019–Aug 2026*

- Led the platform organisation of 45 engineers.
- Built an internal cloud cost dashboard using BigQuery billing exports, Looker Studio and Python scripts; rightsizing and committed-use discounts saved EUR 4.2M per year (31%).
- Contributed to the monolith-to-services migration, moving a Python 2 monolith toward approximately 60 Python 3 and Go services deployed on GKE.
- Established a service template using FastAPI or Go, OpenTelemetry tracing, Prometheus metrics and Terraform modules.
- Improved playback API availability from 99.5% to 99.95%.

Principal Engineer *2015–2019*

- Served as technical lead for playback API design, reliability, and delivery.
- Designed and wrote most of a Go entitlement service handling subscriptions, regions and devices; it handled 120,000 requests per second at peak with p99 latency under 15 ms.
- Used Cassandra with local in-memory caching for the entitlement service.

SELECTED ADDITIONAL EXPERIENCE

Open-source maintainer | Python retry library *2012–2018*

- Maintained and tested a Python retry library using pytest and tox; the project reached approximately 1,500 GitHub stars. Handed over maintenance in 2018.

Co-inventor | Granted German patent *2013*

- Co-inventor of a method for adaptive GPS sampling in vehicle fleets, using driving context to reduce GPS data volume.
- Technologies: C++, statistics.